

theta-Expression

The theta expression constructs the binding for a named schema. When a schema S has components x and y , $\text{theta } S$ is the pair (x, y) viewed as a named record. The primed form $\text{theta } S'$ is the after-state binding.

[*Item*]

Store
$\text{items} : \mathbb{P} \text{ Item}$

A read-only query that returns a snapshot of the current state uses the theta expression to package the before-state as a single output value.

SnapshotStore
$\exists \text{Store}$
$\text{snapshot!} : \text{Store}$
$\text{snapshot!} = \theta \text{Store}$

The Xi-style frame condition $\theta S = \theta S'$ asserts that none of the schema components changed. This is the definition of Xi S expressed using theta directly.

StoreFrame
$\exists \text{Store}$
$\theta \text{Store} = \theta \text{Store}'$